Model Report: *Synthetic Media Detection Engine*

1. Architecture & Technical Pipeline

Verifai utilizes a lightweight Convolutional Neural Network (CNN) optimized for local edge deployment via **ONNX Web Runtime** within a browser extension environment.

- **The Spatial Bottleneck vs. The Power of FFT:** Standard spatial CNNs fail to separate synthetic textures from natural images, capping accuracy at 84.00%. Verifai bypasses this limit using a parallel Spectral Gating Module powered by Fast Fourier Transform (FFT). By converting spatial maps into the frequency domain, hidden generative upsampling artifacts become quantifiable spikes, boosting validation accuracy to 97.00%.
- **Edge Preprocessing:** Inputs are downsampled to 32×32 RGB tensors, normalized, and processed utilizing an optimized NCHW memory format. This compact structure allows the dual-path pipeline to execute locally within a sub-3MB compilation footprint, eliminating host memory drag.

2. Validation & Security Rigor

We engineered the local runtime sandbox with production-grade backend API defenses:

- **Contract Integrity:** We enforce strict type validation schemas via **Pydantic**. Output confidence metrics are programmatically clamped to a rigid [0.0, 1.0] range to neutralize downstream overflow or boundary exploits.
- **Infrastructure Defense (SSRF & DoS):** Incoming image URLs are programmatically sanitized. The extension resolves destination hostnames locally and blacklists private/loopback IP ranges (127.0.0.0/8, 10.0.0.0/8), blocking internal network reconnaissance. To prevent local memory starvation, we enforce a strict 5MB payload ceiling and rigid MIME-type header checks.
- **Reproducibility:** Training pipelines enforce fixed random seeds and deterministic kernel execution to guarantee stable, auditable model weights.

3. Decision Logic & Trade-offs

Feature Choice	Justification	Trade-off / Impact
Spectral Gating (FFT)	Maps data to the frequency domain to isolate mathematical artifacts left by GAN/Diffusion upsampling layers.	Single-handedly broke the 84.16% spatial bottleneck, establishing our verified 96.81% accuracy benchmark over 100,000 CIFAKE assets.
Probability Bucketing	Rejects error-prone binary labels using clear confidence bands: [0.00-0.25] Fake, [0.26-0.75] Unsure, and [0.76-1.00] Real.	Drastically limits false-positive errors near the decision boundary by transparently routing ambiguous assets to human-in-the-loop loops.
Minimal Augmentation	Prioritizes consistent detection of baseline synthetic signatures.	Lower robustness to extreme rotation or heavy JPEG compression.

4. Limitations & Challenges

Verifai's boundaries are defined by its core choice: extreme edge efficiency. The 32x32 tensor bottleneck prevents the model from auditing microscopic forensic flaws like asymmetrical iris geometry or high-resolution facial blending boundaries. Furthermore, if an image undergoes heavy "adversarial scrubbing", such as aggressive downsampling, multi-generational compression, or a physical print-and-scan cycle, the high-frequency artifacts can be smoothed away, degrading the diagnostic utility of the FFT module.